

CEM SARP TAKIM

İstanbul, Türkiye • +90 553 554 9985 • cemsarptakim@gmail.com
cemsarptakim.com • github.com/CemSarp • linkedin.com/in/cem-sarp-takim

SUMMARY

Recent Computer Science graduate (CGPA 3.54) working across AI/ML and full-stack development. Hands-on with deep learning (PyTorch, LoRA) and classical ML in Python and R, plus web and mobile apps on Django REST, Spring Boot, and React. Applied computer-vision models to real infrared imagery during an internship at ASELSAN. Seeking applied AI/ML or software engineering roles.

SKILLS

Languages: Python, R, Java, C++, JavaScript, SQL, PHP

ML & Data Science: PyTorch, NumPy, LoRA, random forests (ranger), EDA, model evaluation & validation

Web & Backend: Django REST Framework, REST API design, JWT authentication, React, React Native, CSS

Databases: PostgreSQL, MySQL, MongoDB, Firebase Firestore

Tools & Platforms: Git, Linux, Vercel, Jupyter, RStudio, Postman

Spoken Languages: English (C1, IELTS: 8.0), Turkish (native), German (A2)

EDUCATION

Sabancı University — İstanbul, Türkiye

Graduated June 2026

BSc Computer Science & Engineering; Minor in Business Analytics | 50% merit scholarship

CGPA: 3.54 / 4.00

Relevant coursework: Machine Learning, Artificial Intelligence, Natural Language Processing, Data Science, Statistical Modelling, Linear Algebra

EXPERIENCE

ASELSAN — Image Processing Intern

06/2025 – 07/2025

Ankara, Türkiye

- Contributed to an AI object-detection pipeline for infrared (IR) surveillance imagery within the Electro-Optical Systems (MEOS) Image Processing team.
- Evaluated open-vocabulary models (OWL-V2, Florence-2) for zero-shot detection of unseen classes against an in-house YOLO baseline, building a benchmarking workflow on custom datasets to quantify the trade-off between inference speed and generalization.
- Prototyped integration of the selected models into an in-house GUI for annotating and verifying IR footage, adding near real-time, model-assisted detection.
- Reduced annotation time by ~15% in internal trials, making the team's shift toward semi-supervised labeling workflows more practical.

IAS (Industrial Application Software) — ERP Consultancy Intern

04/2025 – 05/2025

İzmir, Türkiye

- Contributed to ERP implementation and customization in the caniasERP consultancy department, training on the Project Management (PRJ), System Management (SYS), and Development (DEV) modules and the proprietary TROIA development language.
- Built a company vehicle-management application end to end; database tables, classes, dialogs, and transactions, wiring backend logic to frontend GUI elements.
- Learned client-oriented ERP customization and system configuration, bridging technical implementation with business needs.

BNT PRO — Network Systems Intern

07/2024 – 08/2024

İstanbul, Türkiye

- Supported the Network & Security team's daily operations, including system configuration, troubleshooting, and documentation.
- Helped run cybersecurity fundamentals; log monitoring, user access control, and basic vulnerability scans.
- Shadowed senior engineers through system audits and security assessments, learning enterprise-infrastructure best practices.

PROJECTS

Narrative-Aware Word Sense Plausibility Regressor — SemEval-2026 Task 5

NLP / Deep Learning • Python, PyTorch, DeBERTa-v3, LoRA

- Designed the core model architecture for a 5-person SemEval-2026 Task 5 system that rates how plausible a word's meaning is within an ambiguous short story, scoring each candidate sense on a continuous 1–5 scale.
- Built the QuadPoolRegressor on DeBERTa-v3-large by introducing narrative boundary tokens ([PRE], [HOM], [END]) and specialized pooling layers that separately encode pre-context, ambiguity context, and story ending information.
- Trained with LoRA across a three-stage curriculum (WiC-TSV warm-up, regression, pairwise-ranking); the system reached 0.714 Spearman and 0.825 Acc@SD on the AmbiStory dev set, performing competitively with baseline and reference systems reported for the task.

Label & Co. — Web-Based Supply Chain Simulation Game (Graduation Project)

Full-Stack / Simulation • React, Vercel, Firebase Firestore

- Built a 20-round supply chain simulation game where industrial-engineering students manage four connected roles: Sales, Production, Purchasing, and Logistics. Each with its own decisions and KPIs.
- Implemented the core business logic: demand forecasting, inventory and production planning, purchasing optimization, logistics planning, and a random crisis system that dynamically changes conditions each round.
- Deployed as a React/Tailwind app on Vercel with serverless endpoints; persisted multiplayer sessions in Firebase Firestore and moved the instructor PIN and Firebase config server-side via environment variables.

Pump it Up: Data Mining the Water Table — DrivenData Competition

Machine Learning / R • ranger, caret, tidyverse

- Built an end-to-end R pipeline covering exploratory analysis, data cleaning, feature engineering, model tuning, and leaderboard submission.
- Engineered features from noisy real-world data: converted placeholder strings into explicit missing values, reduced high-cardinality categorical features (30k+ unique values) into statistically meaningful groups, flagged and median-imputed invalid geocoordinates, and derived pump age.
- Tuned a ranger random forest with 5-fold cross-validated grid search over mtry and min.node.size; achieving 0.8205 accuracy on the public DrivenData leaderboard.

SoleVault — Full-Stack E-Commerce Platform

Full-Stack Web • Django REST Framework, React, PostgreSQL, JWT

- Built a full-stack sneaker marketplace with a Django REST backend and a React (Vite) frontend, covering registration, login, profile management, product browsing, and reviews.
- Implemented JWT-based authentication with short-lived access tokens, refresh-token rotation, blacklisting on logout, and route-level access protection.
- Designed REST APIs and PostgreSQL schemas supporting authentication, user profiles, product catalogs, and review workflows.

CSR Communication in Disaster Relief — PURE Research Program

Research • Text classification, social-media data analysis

- Analyzed thousands of tweets from 60 companies posted during the 2023 Kahramanmaraş earthquake to study how the public engaged with corporate disaster-response messaging.
- Classified posts by message type and compared engagement across categories, finding emotional storytelling and visual content to be most strongly associated with higher public trust and engagement.

Real-Time Mobile Messaging App

Full-Stack Mobile • React Native, Java, MongoDB, JWT, WebSockets

- Built a real-time mobile chat app with a React Native (TypeScript) frontend and a Spring Boot backend, supporting direct messages, friend requests, and group chats.
- Implemented live messaging over WebSockets, while exposing REST APIs for authentication, social features, group management, and message persistence in MongoDB.
- Secured the API with Spring Security and JWT. Implemented automatic JWT refresh flows to maintain authenticated sessions without requiring users to re-login.

LEADERSHIP & ACTIVITIES

- **Volunteer Mentor, Civil Involvement Project (CIP101):** Mentored primary-school students from under-resourced areas near Sabancı University, running workshops on communication, creativity, and confidence.
- **Organizer, CAKUF (Cakabey National Forum):** Event planning, logistics, and team coordination.
- **Türkiye Zeka Vakfı:** Top-10 group placement in an interscholastic competition.